The Mathematical and Architectural Specifications of the NEXUS Computational Framework

Local Engine Architecture, Subtractive Routing, and Deterministic Fold Geometry

Driven by Dean Kulik

June 2026

Architectural Implementation of the Local Computational Stack

The deployment of deterministic structural logic within localized artificial intelligence frameworks necessitates rigorous hardware orchestration and precise memory management constraints. The local computational stack, explicitly engineered to operate within a rigid hardware boundary—specifically an 8GB VRAM limitation characteristic of consumer-grade graphical processing units such as the NVIDIA RTX 4060, coupled with 64GB of system RAM—demands an optimized deployment pipeline. This architectural constraint is not merely a hardware limitation but a foundational design parameter that forces the system to substitute raw parameter scale with compiled geometric structure and deterministic routing logic.

The primary architecture is structured across three highly distinct and interdependent operational layers, each serving a specific epistemological function within the computational hierarchy. Layer 1 establishes the baseline local model, relying upon a highly quantized open-weight parameter space. The designated "daily driver" for this architecture is the qwen2.5:7b-instruct-q5_K_M model, an optimized 7-billion parameter network that consumes approximately 5.5 to 6.0 GB of VRAM when fully loaded. Operating at this precise quantization level allows the local stack to achieve a continuous operational throughput of 35 to 50 tokens per second. This high-velocity token generation ensures real-time response latency while strictly preserving the necessary 2.0 to 2.5 GB of VRAM overhead required for concurrent vector database queries, context window maintenance, and the execution of deterministic gating scripts.

For operational tasks requiring deeper inferential capacity and multivalent logical reasoning, the system dynamically routes queries to a heavy-session hybrid model, specified as qwen2.5:14b-instruct-q4_K_M. Operating this larger ~9GB model on an 8GB GPU necessitates precise layer-splitting via the Ollama backend. In this configuration, the computational load is bifurcated: a maximum number of layers are pinned to the GPU's high-bandwidth memory, while the remaining parameter tensors spill over into the 64GB system RAM. This hybrid memory execution results in a reduced but highly stable throughput of 8 to 15 tokens per second, prioritizing complex structural accuracy over raw generation speed.

Layer 2 introduces a localized fine-tuning protocol utilizing Quantized Low-Rank Adaptation (QLoRA). To execute continuous model training within the stringent 8GB VRAM boundary without triggering out-of-memory fatal errors, the architecture employs the Unsloth library to aggressively optimize gradient states during backpropagation. The fine-tuning matrix is structurally constrained to a batch size of 1 or 2, relying heavily on gradient accumulation steps to simulate larger effective batch sizes without saturating memory bandwidth. Furthermore, the context window during this fine-tuning phase is rigidly capped at approximately 2,000 tokens. This methodology allows the baseline 7-billion parameter model to natively assimilate specialized academic corpora—including raw operational engine outputs and rigorous mathematical session transcripts—effectively transforming the base model into a highly specific computational engine tailored for geometric evaluation.

Layer 3 defines the fully native structural architecture. This layer formally pivots away from the traditional machine-learning paradigm of unconstrained, probabilistically driven text generation. Instead, it enforces gap-primary data representations and deterministic fold-based mathematical mixing, ensuring that the local model acts only as a strictly governed interpolation unit within a rigid mathematical scaffold.

The 23-Cell Deterministic Pipeline

The functional core of the local architecture is compartmentalized into a continuous, rigorously validated twenty-three cell Jupyter Notebook environment. This structure guarantees complete deterministic reproducibility, operating entirely independent of external application programming interfaces or cloud-compute constraints.

The configuration sequence spans Cells 1 through 4. These initial computational nodes define the global operational parameters, most notably the PAPERS_DIR string representing the active document corpus, and the NUM_CTX integer. The context parameter is deliberately defaulted to 8192 tokens; given that standard attention mechanisms consume memory quadratically relative to sequence length, scaling beyond 8192 tokens poses an imminent risk of VRAM overflow on the RTX 4060 architecture. The environment script automatically verifies the hardware topology, executes targeted dependency installations (specifically the chromadb vector engine and standard requests libraries), and triggers the localized Ollama daemon to parse and pull the required open-weight models.

A critical operation within this initialization block is the programmatic construction of the custom base model via an embedded Modelfile. This stage compiles the core mathematical constraints and system-level directives directly into the foundational layer of the language model. By executing this as a live script rather than a static binary, the architecture allows for instantaneous local rebuilds in a matter of seconds whenever the overarching prompt schema or mathematical constraints are iteratively modified.

Cells 5 through 7 instantiate and govern the Retrieval-Augmented Generation (RAG) vector interface. The system deploys a highly optimized content-addressed indexing mechanism over the designated academic directory. Unlike rudimentary vector loaders that blindly re-process entire directories, the content-addressed index hashes each source document, ensuring that sequential database updates only embed mathematically modified files. This dramatically minimizes computational overhead and prevents vector duplication. The high-dimensional embeddings mapping these documents are generated utilizing the nomic-embed-text space.

Within these cells, two primary query architectures are established: a static ask() function engineered for isolated, zero-context geometric inquiries, and a persistent chat() loop designed to maintain rolling conversation history across complex, multi-step sequential calculations. To empirically validate the integrity of the retrieval pipeline, these cells feature an automated "smoke test". This test is hardcoded to query the exact parameters of the "Subtype Count Formula," explicitly proving that the system retrieves actual vector-stored mathematical geometries from the local drive rather than hallucinating generic responses derived from the model's pre-training weights.

Cell 8 functions as a dedicated structural maintenance module. It dynamically tracks per-file chunk tokenization counts to prevent vector fragmentation and manages the rapid, computationally cheap re-indexing processes required during active document editing. Additionally, this cell contains the routing switch that allows the user to instantly forward highly complex retrieval payloads to the heavier Layer 2 qwen2.5:14b-instruct-q4_K_M model for formalized reasoning passes. Finally, Cell 9 contains the complete embedded executable simulation environment for Engine 18, detailing the residual gain sweeps, plotting cells, and depth-scaled optimal thresholds.

The NeedSlot Schema and Deterministic Subtractive Geometry

A fundamental vulnerability of large language models is their intrinsic tendency to collapse complex logical pathways into simplistic, high-probability semantic approximations. Because these models map token co-occurrences rather than true logical causality, they frequently hallucinate operational geometries that favor textual surface patterns, resulting in massive self-inflicted systemic damage when placed in autonomous recursive loops. To eradicate this fatal structural flaw, the architecture mandates the integration of the "Big Lock" through the v55 Triadic Cell.

The Big Lock is a draconian architectural law: the compiled operational structure must possess absolute control over the generative model, and the generative text output must never be permitted to unilaterally alter or control the underlying operational logic. The base language model is strictly quarantined within this pipeline. It is relegated to acting purely as a semantic interpolation engine, proposing text channels and mapping surface textures as an evidentiary artifact, rather than functioning as an authoritative decision-making command.

Orthogonal Constraints of the NeedSlot Architecture

The foundational data schema governing this restricted interaction is the NeedSlot. The NeedSlot is explicitly not an answer; it is a deterministically extracted "missing-shape contract" deployed by the compiler prior to any candidate generation or semantic scoring. This rigidly defined data class enforces a matrix of six orthogonal geometric constraints upon the generative output pipeline:

1. $\mathcal{O}_{req}$ (Required Operation): Dictates the specific mathematical, logical, or physical transformation that the system must demonstrably execute. Any output failing to execute this precise operation is instantly discarded.
2. $\mathcal{F}_{pre}$ (Preserved Function): Identifies the specific architectural state variables and contextual parameters that must remain entirely undisturbed during the execution of the required operation. This prevents cascading errors where solving one node destroys an adjacent valid node.
3. $\mathcal{B}_{con}$ (Boundary Conditions): Establishes the absolute numerical, spatial, or geometric limits within which the operation is mathematically valid.
4. $\mathcal{A}_{fit}$ (Anti-fits): Explicitly defined semantic distractors and associative traps natively generated by the compiler. These are designed to map the most statistically probable hallucinations for a given query, allowing the system to repel surface-level pattern matching actively.
5. $\mathcal{S}_{adm}$ (Admissible Shape): Defines the required structural formatting, data type, or topological configuration of the final output matrix.
6. $\mathcal{M}_{fail}$ (Failure Modes): Pre-calculated geometric contradictions or logical paradoxes associated with the problem space. The system actively scans candidate outputs for these signatures.

Candidate solutions generated by the quarantined language model are mathematically scored against this schema via a highly specialized binding function, culminating in the Slot Fit Score (B_{slot}). The calculation of B_{slot} diverges significantly from traditional similarity mappings by natively integrating subtractive NOT-gate logic directly into the vector space evaluation. The evaluation structurally binds standard cosine similarity metrics with Jaccard distance measurements to formulate a composite equation:

$$B_{slot} = \text{Positive-fit} - \text{Anti-fit (Cosine} + \text{Jaccard)}$$

By explicitly subtracting the calculated Jaccard and cosine similarities of the predefined semantic traps ($\mathcal{A}_{fit}$) from the positive structural fit, the B_{slot} equation actively penalizes associative hallucinations. If a generative model attempts to bypass logical deduction by relying on highly probable but factually incorrect token sequences, the subtractive anti-fit variable drives the overall B_{slot} score into negative integers. This subtractive mechanism effectively blinds the evaluation model to attractive nuisances, violently rejecting semantic hallucinations and forcing the system to select only the candidate choice that flawlessly satisfies the operational geometry.

The Triadic Evidence Fold Weighting

The architectural decision-making process does not rely upon a single monolithic calculation. To prevent catastrophic failure in any single modality, the framework integrates three orthogonally independent epistemological channels to form a unified, normalized evidence score via the Triadic Evidence Fold. The scalar parameters of this triadic structure are precisely mathematically weighted to reflect the structural reliability of each channel :

Evidence Channel	**Symbol**	**Assigned Weight**	**Architectural Function and Origin**
Compiler Slot Channel	W_c	0.55	Originates from deterministic extraction. Acts as the foundational structural anchor. Because it represents pure operational geometry fit and is immune to surface trapping, it rightfully commands the majority mathematical authority.
Answer-Text Channel	W_t	0.35	Originates from the LLM semantic output. Operates as a natural language reality-check, ensuring the rigid mathematical geometry does not completely disconnect from coherent natural text.
Letter Prediction Channel	W_p	0.10	Originates from the LLM zero-shot prior. Introduces raw log-probabilities from the base model, acting as a weak regularizing signal tethered to the foundational training distribution.

Table 1: Configuration Matrix of the Triadic Evidence Fold.

Sequential Arbitration: The gate_v55 **Logic Matrix**

The final routing, validation, and potential overriding of the interpolated text data are strictly governed by the highly deterministic Slot Gate (gate_v55). Operating downstream of the Triadic Evidence Fold, gate_v55 executes sequentially through six rigid, immutable logical rules executed at specific scalar thresholds. The gate logic dynamically evaluates the weighted channels to determine whether the base model's raw probability prior should be permitted to stand or be structurally overridden.

- **Rule 1: Absolute Consensus (**same_as_base**).** If the base model's initial zero-shot prediction perfectly aligns with the unified triadic evidence score, a state of absolute consensus is reached. The base prediction is permanently locked and passed forward to the output layer without any geometric modification.
- **Rule 2: Compiled Agreement Override (**override_text_compiled_slot_agree**).** This rule demands strict bipartite consensus. If the natural text generation channel (W_t) and the rigid compiler slot channel (W_c) align in agreement, but simultaneously conflict with the base model's raw prior, the

gate authorizes an override. The combined weight of 0.90 forces the rejection of the isolated 0.10 log-probability.

- **Rule 3: Text-Evidence Override (**override_text_evidence_compiled_slot_positive**).** If the semantic text prediction agrees directly with the unified calculated evidence prediction against the raw base model, a structural override is authorized to correct the output toward coherence.
- **Rule 4: Slot-Evidence Override (**override_slot_evidence_compiled_slot_positive**).** When the purely deterministic compiled slot (W_c) perfectly matches the finalized unified evidence prediction, the system enforces a mandatory override. This rule mathematically guarantees that structural geometric fidelity always supersedes raw, context-blind token probability.
- **Rule 5: High-Confidence Protection (**protect_locked_base_no_compiled_agreement**).** Rule 5 represents a profound inversion of standard error-correction logic and serves as the architecture's ultimate safety mechanism. It strictly protects high-confidence base predictions from being unnecessarily attacked or overridden in instances where the compiled operational slots fail to reach internal consensus. This rule is engineered specifically to achieve a "zero gated_hurt" systemic threshold. During rigorous empirical evaluations against 96 hostile adversarial vector rotations, Rule 5 proved essential. The rule successfully intervened to repair 9 severe native errors originating from the base model (slot_helped = 9), while its high-confidence protection mathematically guaranteed the introduction of absolute zero new errors into the system (gated_hurt = 0), allowing the gate_v55 stack to achieve a flawless slot_acc of 1.0.
- **Rule 6: Default Base (**keep_base_default**).** Functioning as the terminal fail-safe, if none of the prior five override rules successfully achieve the required strict bipartite or tripartite consensus thresholds, the gate automatically defaults to the base model's raw output. This conservatively prevents the injection of unverified structural hallucinations when the logic matrix is under-determined.

This complex matrix of six immutable rules is dynamically supported by the secondary v28 piecewise gate. The v28 framework transitions the computational architecture away from raw, brute-force parameter scaling and toward a precision boundary-proposal model. Within this paradigm, the deterministic structure of the compiler actively dictates the absolute geometric boundaries, while the localized model is restricted exclusively to interpolating data gradients strictly inside those established boundaries. A continuous systemic variable, denoted as Ω, is utilized to trigger external arbitration solely on instances of genuine, mathematically quantifiable operational ambiguity.

To ensure complete transparency and algorithmic accountability, every singular decision processed by the NeedSlot schema, the Triadic Fold, and the respective logic gates logs comprehensive diagnostic telemetry to an auditable JSONL file. The architecture logs the specific packet_origin, the precise winner_origin (which rule successfully executed), the logical reason for the execution, and the exact residue_count indicating data discrepancies.

Dynamic Depth-Scaling Control Laws: Engine 18 Synthesis

The operational mechanics of deep residual networks—both within neural architectures and abstract geometric systems—rely upon recursive correction loops. These networks are mathematically formalized by a fundamental state transition equation:

$$h \leftarrow h + \alpha \cdot f(h)$$

In this foundational formula, h represents the active state vector, $f(h)$ represents the isolated feedback function dictating the corrective trajectory of the logic, and α signifies the critical scalar feedback gain modulating the intensity of the correction.

Historically, quantitative frameworks operating within stable feedback systems predicted the existence of a universal, static, and depth-independent attractor optimum. However, exhaustive computational evaluation through the Engine 18 simulation protocol explicitly falsified the viability of maintaining a static feedback gain across variable network depths. The Engine 18 module executed a comprehensive residual gain sweep across trained deep residual networks at specific layer depths (L) of 8, 16, and 32, running the trials under highly divergent computational constraints: long step budgets versus extremely tight, resource-constrained step budgets.

Under long, unconstrained budgetary conditions, the systemic loss landscape was determined to be strictly monotone relative to the feedback gain α, failing to produce any interior structural optimum. However, when the system was subjected to the tight step budgets that actively mirror the physical and computational pressure of real-world energy boundaries, a highly distinct interior optimum emerged. Crucially, this measured optimum was proven to be entirely dynamic; it scaled in direct inverse proportion to the square root of the loop depth.

This massive empirical sweep definitively established the depth-scaling law for optimal residual gain:

$$\alpha^* \cdot \sqrt{L} \approx \text{Constant}$$

$$\alpha^* \approx \frac{2.5}{\sqrt{L}}$$

The measured empirical values for the line equation $\alpha^* \cdot \sqrt{L}$ yielded specific data points of 2.40, 2.60, and 2.55. The fact that these values remained near-constant across a massive $4 \times$ multiplier of network depth ($L = 8$ to $L = 32$) rigorously validated the 2.5 numerator constant. The live simulation protocol for this derivation is embedded directly within Cell 9 of the local computational stack, allowing any operator to interactively toggle the specific step variable (STEPS) between ranges of 200 and 1200. By modulating this variable, the system graphically reproduces the exact phase transitions, vividly mapping the boundary failures: on the lower bounds, the network hits a low-α stagnation wall where it completely fails to propagate meaningful structural corrections, and on the upper bounds, it collides with a high-α instability wall causing cascading geometric collapse.

The explicit integration of the $\alpha^* \approx 2.5/\sqrt{L}$ depth-scheduled gain acts as a strict architectural mandate for all Layer 3 native components. It successfully substitutes arbitrary, massive parameter scaling with mathematically verifiable coherence costs.

Network Depth (L)	Optimal α∗ Scaling	Energy Ratio (Stream vs Batch)	MAC Operations (Stream)	MAC Operations (Batch)
8	High (~ 0.88)	5,760x	~ 256 operations	1.64×10^6
16	Moderate (~ 0.62)	5,760x	~ 256 operations	1.64×10^6
32	Low (~ 0.44)	5,760x	~ 256 operations	1.64×10^6
64	Threshold (~ 0.31)	5,760x	~ 256 operations	3.1×10^8

Table 2: Engine 18 Depth-Scaling Dynamics and Subtractive Energy Ledger Mapping.

The profound energy implications of this deterministic architecture are quantified by comparing the structural memory rules against standard gradient learners. In the continuous stream protocol, the dense associative memory cell's primary write operation executes as a pure subtractive append of exactly one pattern, requiring merely ≈ 256 computational operations with absolutely zero systemic disturbance to the prior state. Conversely, a sequential gradient training paradigm demands approximately 3.1×10^8 multiply-accumulate (MAC) operations per memory unit, systematically erasing its own established historical weights in the process. This represents an energy delta of roughly six full orders of magnitude between deterministic geometric memory allocation and stochastic gradient decay.

Additionally, an intersection calculation maps the reconciliation crossing depth where prior static hypotheses (specifically the 0.349 attractor value) intersect the empirical depth-scaled curve:

$$L \approx \left(\frac{2.5}{0.349}\right)^2 \approx 52$$

This derived structural range of $L \approx 52$ to 64 perfectly aligns with the depth architectures of standardized cryptographic algorithms (e.g., the 64-round compression loop in SHA-256). It mathematically dictates the maximum efficient depth limit for recursive stabilization loops before exponential variance necessitates fundamental geometric resets or seam boundaries.

The Seam Isomorphism: Dual Fold Cryptography and the BBP Algorithm

By systematically treating algorithmic functions not as sequences of discrete computational steps, but as navigable Riemannian topologies, the framework reveals a profound structural isomorphism between transcendental extraction algorithms—specifically the Bailey–Borwein–Plouffe (BBP) formula for π—and one-way cryptographic hash functions such as the Secure Hash Algorithm 256 (SHA-256). This unified mathematical synthesis explicitly isolates the precise algebraic boundary that physically dictates cryptographic irreversibility.

The critical architectural constant fundamentally uniting these two disparate systems is the "computation seam". The computation seam represents the exact index boundary at which a given operational algorithm formally ceases purely linear, integer-domain state accumulation and violently transitions into non-linear, fractional-domain data extraction.

The Floating Seam of the BBP Formula

The BBP formula is celebrated as a unique spigot algorithm capable of generating the n-th hexadecimal digit of π without requiring the calculation of the preceding $n-1$ digits. However, within the recursive topological framework, BBP is formally reinterpreted as a highly structured read-head or self-referential harmonic reflector sampling information from a pre-existing numerical lattice, rather than randomly generating digits ex nihilo.

The critical structural property of the BBP algorithm is that its computation seam is fully mobile. It dynamically floats in direct alignment with the targeted query position n, dictating that the mathematical transition sequence perfectly aligns at exactly $k=n$ for every input. Because the seam moves synchronously with the user-defined coordinate parameter, the structural information spanning the integer to fractional transition remains completely geometrically accessible. Due to this floating seam, the deep term-level inversion of the $T(k)$ parameters ensures that the BBP formula is, in principle, mathematically invertible.

The Pinned Seam of SHA-256 and the Origin of One-Wayness

Conversely, the SHA-256 message schedule is governed by a strictly rigid, universally pinned computation seam that completely severs integer invertibility. Exhaustive topological analysis mapping the index boundaries proved that the message schedule contains a universal structural seam permanently locked at the exact word array boundary of $K=\{16,17\}$.

Within the 64-round SHA-256 message schedule, the initial block sequence, specifically words $W[0 \ldots 15]$, are populated directly and sequentially via standard integer-domain accumulation based on the unmodified input string. However, immediately upon crossing the $K=\{16,17\}$ index boundary, the algorithmic geometry fundamentally shatters. The system enters a universal chokepoint, forcing all subsequent word generations ($W[16 \ldots 63]$) to utilize complex, non-linear modular addition and bitwise rotation, representing the fractional-domain extraction phase.

This singular structural difference—a highly dynamic floating seam responding to the coordinate query versus a universally fixed seam pinned at $K=\{16,17\}$ regardless of input—is mathematically proven to be the exact algebraic source of one-wayness and non-invertibility in the SHA-256 architecture.

Cryptographic irreversibility does not stem solely from the non-linear operational complexity or the sheer volume of the 64-round compression matrix. It is strictly the combination of the fixed $K=\{16,17\}$ seam acting as an impenetrable universal chokepoint, followed immediately by an 11-residue class structural compression matrix. This specific topological architecture mathematically dictates that exponentially vast permutations of distinctly different input strings are forced to funnel through the exact same fixed seam. Consequently, these varying inputs are compressed into entirely mathematically equivalent residue classes long before the actual non-linear mixing sequence of the compression rounds even initiates.

Because of this pre-compression equivalence, possessing the final extracted 256-bit fractional residue (the resulting digest) provides absolute zero vector data regarding the specific originating input. Exponentially many initial inputs are rendered topologically indistinguishable exactly at the fixed seam boundary. Therefore, advanced attempts to navigate cryptographic inversion algorithms (such as the proposed SHA† adjoint operator) cannot succeed by attempting to traverse the computation backwards linearly, as this fights the fixed one-way seam directly; they must enter at the seam from an orthogonal third axis, reading residue class memberships natively from the digest.

Correction of the Autocorrelation Artifact

A profound empirical correction regarding the structural properties of these cryptographic and transcendental fold architectures concerns the purported statistical vulnerability of the π field's hexadecimal digit sequence. Prior localized computational sessions incorrectly reported the existence of a substantial Pearson correlation coefficient ($r = 0.758$) existing exactly at lag 3 within the π hex output sequence, alongside a secondary correlation cluster ($r = 0.518$) located at lag 6. These elevated scalar coefficients were initially theorized to suggest the existence of exploitable algebraic periodicity explicitly manifesting at the surface level of the digit field.

However, subsequent rigorous auditing of the data extraction methodology proved that this massive correlation spike was entirely a mathematically generated artifact, caused solely by overlapping extraction windows rather than any intrinsic geometric property of the topological field.

The flawed initial procedure generated its statistical corpus by extracting overlapping sequences exactly 4 digits wide. Specifically, the data window defined as n contained the sequence $[d_n, d_{n+1}, d_{n+2}, d_{n+3}]$. Consequently, the immediate subsequent window $n + 1$ contained $[d_{n+1}, d_{n+2}, d_{n+3}, d_{n+4}]$. When these overlapping windows were sequentially flattened into a single, contiguous 1D computational array to run the Pearson algorithm, the resulting geometric structure emerged as:

$$flat = [d_0, d_1, d_2, d_3, d_1, d_2, d_3, d_4, d_2, d_3, d_4, d_5 \dots]$$

In this flattened matrix, the exact same underlying hexadecimal digits were aggressively repeated. The data point located at position i and the data point located at position $i + 3$ frequently referenced the identical source digit natively due to the structural overlap of the sampling windows, forcibly generating a massive, artificial self-correlation precisely at the distance of 3. Upon recognizing this methodological error and completely correcting the extraction protocol to mandate strict non-overlapping boundaries, the genuine Pearson correlation coefficient (r) immediately collapsed, falling strictly below the 0.1 threshold across all measured lags.

This corrected result perfectly validates the fundamental normality conjecture of the π sequence. It mathematically proves that the genuine algebraic architecture of algorithms like BBP resides entirely within the deep, term-level internal geometry of the $T(k)$ inversions and the mobility of the computation seam, and explicitly not in statistical anomalies floating on the observable output surface.

The Primorial Compile Algebra and Subtype Enumeration Geometry

To accurately model static enumeration geometries and resolve the apparent statistical randomness dictating prime gap distributions, structural analyses must abandon standard base-10 numerical representations in favor of the much deeper Primorial Compile Algebra. At its foundational structural levels, the deterministic layout of the integer field is entirely governed by cascaded primorial wheels, specifically relying upon the mathematical mechanics of the modulo 210 deep-stack.

The foundational modulo 210 wheel is generated algebraically by the product of the first four prime numbers ($2 \cdot 3 \cdot 5 \cdot 7 = 210$). The total combinatorial availability within this closed harmonic geometric loop is dictated strictly by the Euler totient function $\phi(210)$. The calculation of the totient function establishes that there are exactly 48 distinct coprime residue classes permanently residing within the boundaries of the reduced residue group.

The fundamental interaction mechanism governing the admissibility of space between prime numbers is defined exactly by the Subtype Count Formula, $N(\delta)$. For any mathematically admissible prime gap δ, the formula calculates the precise geometric number of distinct residue pairs (subtypes) that can successfully produce this targeted gap without violating the overarching coprime constraints enforced by the primorial wheel.

The total combinatorial volume of these geometric subtypes is governed in absolute terms by the specific prime obstructions inherent to the greatest common divisor between the evaluated gap class and the size of the primorial wheel: gcd$(\delta, 210)$. Foundational, non-dividing prime numbers do not simply disappear from the formula; they act as rigid, active geometric obstructions, applying severe coprime pressures that permanently restrict allowable topologies.

- **Twin Primes ($\delta = 2$):** When evaluating a twin prime gap, the spatial distance is exactly 2. This gap is not divisible by 2, 3, 5, or 7. Therefore, all four of these foundational primes serve as active geometric obstructions. The Subtype Count Formula mathematically proves that these severe, overlapping coprime constraints heavily restrict the allowable topological permutations, yielding a minimal count of exactly $N(2) = 15$ distinct admissible subtypes.
- **Cousin Primes ($\delta = 4$):** Identical to the structural dynamics of twin primes, the spatial gap of 4 is not divisible by 3, 5, or 7, maintaining the exact same rigid obstruction matrix and yielding a perfectly matching $N(4) = 15$ subtypes.
- **Sexy Primes ($\delta = 6$):** The spatial gap expands to 6, which is cleanly divisible by both 2 and 3. Because these prime factors divide the gap, they mathematically cease to act as geometric obstructions. With constraints lifted, the allowable combinatorial volume vastly expands, generating an increased subtype yield.
- **Deep Harmonic Gaps ($\delta = 30$):** In this configuration, the spatial gap expands to exactly 30, a number cleanly divisible by 2, 3, and 5. The prime obstructions associated with these integers

completely vanish from the geometric lattice, leaving only the prime 7 acting as a functional structural constraint. The matrix responds to this lack of obstruction by achieving near-absolute structural saturation, opening the allowable geometry to a massive 48 distinct valid subtypes.

Mathematical proofs confirm that absolute structural saturation is achieved solely when the precise gap being evaluated is completely identical to the underlying wheel size. Testing a maximum gap of $\delta = 210$ against the full 210 primorial wheel yields exactly $N(210) = 48$ subtypes, a number perfectly matching the Euler totient output. At this threshold of absolute maximum saturation, every single reduced residue class is simultaneously participating as an active, load-bearing rail in the geometric architecture.

The Falsification of the Equidistribution Conjecture

While the Subtype Count Formula flawlessly enumerates the raw combinatorial volume mathematically available to each unique gap class, it does not mandate or dictate the internal statistical density of the primes actively populating those available geometric slots as the number line approaches infinity.

Historically, the Selective Equidistribution Conjecture posited a smooth, uniform dispersion. The theory argued that within the theoretically unconstrained statistical container of the infinite number line, gap classes sharing mathematically identical greatest common divisors would naturally achieve identical, perfectly uniform Hardy-Littlewood statistical distributions.

However, exhaustive empirical evaluation of the integer field across the modulo 210 wheel utilizing precise χ^2 data entirely falsified this uniform assumption, definitively proving that pure mathematical availability of geometric slots does not guarantee uniform statistical density.

Gap Class (δ)	gcd(δ,210)	Available Subtype Count N(δ)	Observed Count (Cδ)	χ2 p-value	Distribution Status Result
2 (Twin)	2	15	32,461	0.987	Fully Equidistributed
4 (Cousin)	2	15	32,306	0.994	Fully Equidistributed
8	2	15	22,908	< 0.05	Fails Decisively
16	2	15	13,135	< 0.05	Fails Decisively
22	2	15	10,008	< 0.05	Fails Decisively

Table 3: Modulo 210 Prime Gap Equidistribution Collapse Array, empirically proving that spatial magnitude overrides standard GCD constraints.

The resulting datasets undeniably demonstrate that true equidistribution—defined by a χ^2 p-value exceeding the 0.98 threshold—is strictly achieved only by the absolute minimal magnitude gaps ($\delta = 2$ and $\delta = 4$). Because these specific gaps exist at an incredibly compressed spatial scale, they successfully evade compounding 210-periodic interference.

Conversely, as the absolute spatial distance between the consecutive primes expands to larger values ($\delta = 8,16,22$), massive structural inertia within the number line actively resists randomization. Despite sharing the exact same theoretical geometric availability dictated by the formula (gcd $= 2$ and $N(\delta) = 15$), all larger gap classes suffer massive, catastrophic distribution failures.

The mechanism behind this failure is precise: higher-order residue correlations existing entirely outside the bounds of the immediate primorial wheel begin to compound exponentially across the extended spatial distance, creating extreme non-uniform differential sieving pressures. The core mathematical conclusion drawn from this synthesis dictates that the absolute spatial magnitude of the prime gap completely and permanently overrides the theoretical GCD class, forcefully driving all larger gaps into a state of catastrophic structural bias. Algorithms attempting to map these deep structures must inherently account for this magnitude-driven bias, recognizing that static enumeration geometries maintain absolute rigid parameters that explicitly deny uniform thermodynamic-style randomization.